Kaleb Coleman

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SUMMARY

M.S. robotics student building reproducible simulations for two-arm manipulation and quadrotor control. Background in machine learning, data engineering, and enterprise cybersecurity.

EDUCATION

M.S., Artificial Intelligence Engineering (Robotics) Expected May 2028
Arizona State University, Mesa, AZ

B.S., Data Science; Cybersecurity Minor May 2026
Northern Arizona University, Flagstaff, AZ

PROFESSIONAL EXPERIENCE

Information Systems Security Analyst Intern | Gila River Indian Community Jun 2026 - Present

- Configured and managed endpoint and mobile security tools, including next-generation antivirus, across an enterprise device fleet.
- Monitored network detection and response and SaaS threat intelligence platforms; investigated anomalies and potential security risks.
- Supported email security and phishing defense, and used secure remote access to troubleshoot security issues across the organization.
- Worked with highly confidential information in a tribal government environment supporting government agency operations.

TECHNICAL SKILLS

Robotics: NVIDIA Isaac Sim, MuJoCo, inverse kinematics, model predictive control, system identification

Programming / Data: Python, SQL, R, C, pandas, NumPy, PostgreSQL, SQLite, API ingestion

ML / Tools: PyTorch, scikit-learn, computer vision, Linux, Git, Docker, GPU computing, Slurm

Cybersecurity: endpoint security, network threat monitoring, SaaS security, phishing defense, secure remote support

PROJECTS

Two-Arm Manipulation in NVIDIA Isaac Sim | RISE Lab 2026 - Present

- Developed a simulation setup with opposing UR5e arms and RG2 grippers manipulating a shared bar; added collision validation, joint-frame corrections, object-space goals, and per-arm inverse kinematics.
- Validated a 26-second coordinated-motion simulation with no unexpected collisions and 0.135 mm maximum bar slip.
- Documented scene assets, failure cases, and setup instructions to support reproducible experiments.

Energy-Aware Prescribed-Time Quadrotor Control 2026 - Present

- Developed reproducible comparisons and saved-output baselines for an S500-based MuJoCo test bench studying tracking deadlines, motor lag, and drag.
- Added solver timing diagnostics and contributed to a paper draft; results are simulation-based.

Image Reconstruction and Generative Modeling Spring 2026

- Built PyTorch autoencoder, denoising, VAE, and diffusion experiments on MNIST-family datasets; compared nonlinear reconstruction with PCA baselines using cross-validation.
- Implemented DDPM-style UNet, timestep embeddings, noise scheduling, and reverse sampling; organized reproducible GPU/Slurm runs, isolated outputs, and parameter sweeps.

Cross-Sport Shot Analysis | NAU Capstone 2025 - 2026

- Engineered NBA/NHL ingestion, validation, relational storage, and idempotent upserts across approximately 3 million shot records from 2014-2024.
- Built generalized additive models and a Shot Difficulty Index; visualized shot quality and scoring efficiency and presented the research at NAU UGRADS.

AWARDS & HONORS

NAU ASA DataFest 2026 - Best Visualization Award: Built models, engineered features, and visualized early signals of high healthcare utilization with a team in a 48-hour data science competition.